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Biochar and Soil Physical Properties

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Biochar is considered to be a potential soil amendment. However, its implications for soil physical and hydraulic properties have not been widely discussed. Changes in the soil physical environment influence the numerous services that soils provide. This paper (i) reviewed the impacts of biochar on soil compaction, mechanical, structural, hydraulic, and thermal properties; (ii) discussed factors affecting biochar performance; and (iii) identified research areas. Biochar generally reduces soil bulk density by 3 to 31%, increases porosity by 14 to 64%, and has limited or no effects on penetration resistance. Biochar increases wet aggregate stability by 3 to 226%, improves soil consistency, and has mixed effects on dry soil aggregate stability. It increases available water by 4 to 130%. Saturated hydraulic conductivity decreases in coarse-textured soils, and increases in fine-textured soils following biochar application. Studies on other properties are few but suggest that biochar reduces tensile strength and particle density, alters water infiltration, moderates soil thermal properties, and has minimal effect on soil water repellency. Sandy soils appear to respond more to biochar than clayey soils. Biochar effectiveness increases as the amount of biochar applied increases. A decrease in biochar particle size can increase water retention but may reduce saturated flow. Field-scale and long-term studies assessing all soil physical properties under different scenarios of biochar management are needed. Overall, biochar generally improves the soil physical environment, but long-term field studies are lacking to conclusively ascertain the extent of biochar effects.

Abbreviations: COLE, coefficient of linear extensibility; WDPT, water drop penetration test

Core Ideas

- **Biochar generally improves the soil physical environment**
- **Sandy soils appear to respond more to biochar than clayey soils**
- **Long-term studies are needed to conclusively discern the extent of biochar effects**

Biochar, a product of slow and incomplete combustion of organic materials, is generating interest as a potential soil amendment. Because soil organic C accumulation is central to the improvement in soil properties, addition of C-enriched amendments such a biochar (60–80% C) could enhance soil properties. Biochar technology could be particularly beneficial for soils with low organic C content. Further understanding of the potential soil benefits of biochar for different soil types and management scenarios is needed.

Many papers have reviewed the potential impacts of biochar on soil C sequestration (Lal, 2015; Brassard et al. (2016), greenhouse gas emissions (Brassard et al., 2016), soil fertility (Ding et al., 2016), crop production (Biederman and Harpole, 2013), soil biota (Lehmann et al., 2011), soil chemical properties (Brassard et al., 2016; Ding et al., 2016; Mukherjee and Lal, 2017), and remediation of contaminated soils (Safaei Khorram et al., 2016). However, a review of the implications of biochar application on soil physical and hydraulic properties is lacking. Omondi et al. (2016) reviewed the effects of different types of biochar on five select physical properties (bulk density, porosity, available water content, saturated hydraulic conductivity, and wet aggregate stability). A discussion of all soil physical and hydraulic properties is warranted to better understand the limitations and opportunities of biochar use and management under different soils and biochar management.

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Soil physical and hydraulic properties directly and indirectly influence the services that soils provide. For example, soil physical and hydraulic properties influence macro- and microscale processes including root growth, aeration, compaction, water and nutrient uptake, surface and subsurface water pollution, erosion, and water, heat, and gas fluxes, among others. Thus an improved understanding of how soil physical and hydraulic properties respond to biochar addition is critical to the overall soil performance. The objectives of this review were to (i) synthesize and discuss the impacts of biochar on soil compaction, mechanical, structural, hydraulic, and thermal properties using published studies and (ii) discuss potential factors affecting biochar performance as well as identify research areas.

BIOCHAR AND COMPACTION OR MECHANICAL PROPERTIES

Bulk Density

Soil bulk density is one of the most studied properties following biochar application. Table 1 reports recent studies from 22 soils, which were published in 2016. Omondi et al. (2016) reviewed studies on biochar and soil bulk density prior to 2016. Thus the present paper reports data on soil bulk density only for 2016. Biochar application reduced bulk density by 3 to 31% in 19 out of 22 soils, indicating that bulk density generally decreases with biochar application. On average, bulk density decreased by 12%. The significant decrease in bulk density agrees with Omondi et al. (2016), who reported that biochar application can reduce bulk density by 7.6%.

Bulk density gradually decreased with an increase in the amount of biochar applied (Table 1). As the amount of biochar increased, bulk density decreased linearly (Głąb et al., 2016; Liu et al., 2016a), but in a few cases, it decreased quadratically (Rogovska et al., 2016). The study by Rogovska et al. (2016) found that biochar application rates above 60 Mg ha^{-1} had a smaller effect on reducing bulk density than application rates below 60 Mg ha^{-1}. This literature review also shows that, in a few cases, biochar application at rates <10 Mg ha^{-1} (Usowicz et al., 2016; Xiao et al., 2016) may not significantly reduce bulk density. Furthermore, while application of large amounts of biochar generally reduces bulk density. Pratiwi and Shinogi (2016) found that biochar application even at rates as high as 2% by weight (which is equivalent to about 50 Mg ha^{-1} of biochar) did not affect bulk density. This suggests that biochar effects should be evaluated on a site-specific basis.

Table 1 indicates that biochar application can reduce bulk density in coarse-textured soils more than in fine-textured soils. It reduced bulk density by 14.2% in coarse-textured soils and 9.2% in fine-textured soils. Indeed, the largest decrease (31%) of bulk density in Table 1 was for sand (Liu et al., 2016a). At least two mechanisms could be responsible for the reduction in bulk density after biochar application. First, biochar has a lower bulk density (<0.6 g cm^{-3}) than soil (~1.25 g cm^{-3}). Thus biochar application probably reduces the density of the bulk soil through the mixing or dilution effect. The extent of this effect can be particularly large when the difference in density between materials is large. For example, the effect of biochar for reducing bulk density on clayey soils can be smaller than that on sandy soils because the difference in bulk density between biochar (~0.6 g cm^{-3}) and clayey sols (~1.1 g cm^{-3}) is smaller than that between biochar (~0.6 g cm^{-3}) and sandy soils (~1.5 g cm^{-3}). Second, biochar could also reduce bulk density in the long term by interacting with soil particles and improving aggregation and porosity. The latter requires the monitoring of changes in bulk density through extended periods of time. Most of the studies on biochar were relatively short-term (<4 yr), which may not fully reflect the long-term effects of biochar. In general, biochar application reduces bulk density but the magnitude of these changes can vary with soil and biochar application rate.

Particle Density

Changes in bulk density and particle density after biochar application directly influence soil porosity. Particle density is not commonly measured in routine tests during soil characterization, but it is an important soil property that affects many other soil properties and processes(porosity, particle sedimentation, specific surface area, thermal properties, and others). Most importantly, particle density affects soil porosity. The particle density of biochar materials ranges from 1.5 to about 2 g cm^{-3} (Brewer et al., 2009), whereas the particle density of the soil could range from 2.4 to 2.8 g cm^{-3}, depending on the textural class. In practice, particle density of the soil is often assumed to be constant (2.65 g cm^{-3}) when computing soil porosity. However, changes in soil C concentration could significantly reduce this soil property (Blanco-Canqui et al., 2006). Thus the addition of biochar (>60% C) could induce changes in particle density and thereby affect soil porosity.

Studies specifically measuring biochar effects on soil particle density are few but these studies indicate that biochar application can reduce particle density. In a field study, application of 30 Mg ha^{-1} of wood biochar reduced soil particle density from 2.55 to 2.20 g cm^{-3} (14% decrease) compared with <10 Mg ha^{-1} of wood biochar on a fallow land, but it had no effect on soil particle density on a grassland (Usowicz et al., 2016). In a laboratory study using a loamy sand, Githinji (2014) found that particle density decreased linearly (r^2 = 0.915) with biochar application at rates of 0, 25, 50, 75, and 100% by volume. The particle density values were 2.62 g cm^{-3} for 0%, 2.43 g cm^{-3} for 25%, 2.37 g cm^{-3} for 50%, 2.09 g cm^{-3} for 75%, and 1.60 g cm^{-3} for 100% application rate of biochar, indicating that application of biochar at 100% by volume can reduce particle density by 64%. Similar to the effect on soil bulk density, the decrease in soil particle density with the addition of biochar is attributed to the low particle density of biochar particles. In summary, the few studies indicate that biochar application can reduce the particle density of the soil.

Porosity

The large decrease in soil bulk density and particle density with biochar application can affect soil porosity. Table 1 indicates that biochar application can increase the porosity of the

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Table 1. Impact of biochar application on soil bulk density and total porosity in the upper 15 cm of the soil depth for different soils ranging from sandy to clayey under various management scenarios.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Bulk density	Decrease in bulk density	Increase in porosity	Reference‡
					°C		Mg m^{-3}	%	%	
USA	Silica sand	Laboratory	n/a¶	Mesquite	400	0%	1.62a§		12 to 41	Liu et al. (2016a)
						2%	1.45b	–10		
						4%	1.34c	–17		
						6%	1.28d	–21		
						8%	1.20e	–26		
						10%	1.11f	–31		
Sri Lanka	Sand	Laboratory	6 mo	Rice husk	~600	0%	1.48a			Gamage et al. (2016)
						0.1%	1.39b	–6	7	
						0.5%	1.32c	–12	14	
						1%	1.27c	–17	18	
	Sandy loam					0%	1.41b			
						0.1%	1.31cd	–8	8	
						0.5%	1.28de	–10	11	
						1%	1.24e	–14	13	
Poland	Sandy loam	Field	1.5 yr	Wood	350–400	0 Mg ha^{-1}	1.18a		Increased	Usowicz et al. (2016)
						10 Mg ha^{-1}	1.20a			
						20 Mg ha^{-1}	1.04b	–13		
						30 Mg ha^{-1}	1.00b	–18		
China	Sandy loam	Field	4 yr	Wheat straw	350–550	0 Mg ha^{-1}	1.13a		n/a	Zheng et al. (2016)
						20 Mg ha^{-1}	1.09ab			
						40 Mg ha^{-1}	1.06b	–7		
China	Sandy loam	Field plots (5 by 6 m)	4 yr	Wheat straw	350–500	0, 5, and 10 Mg ha^{-1}	Reduced	–2.9	n/a	Qin et al. (2016)
India	Sandy loam	Field plots	3 yr	Corn stalk, cotton, stalk, red grama stalk, and mesquite	350–400	0 Mg ha^{-1}	~1.42a		Increased	Pandian et al. (2016)
						2.5 Mg ha^{-1}	~1.37b	–4		
						5 Mg ha^{-1}	~1.36b	–4		
Zambia	Sandy loam, loamy sand, and sand	Field	1 yr	Corn cob and rice husk	350	0–4%	Reduced	–3 to -20	2 to 12	Obia et al. (2016)
Austria	Sandy loam	Greenhouse	3 yr	Control	525	0%	1.43f		n/a	Burrell et al. (2016)
				Wheat straw		3%	1.17a	–22		
				Woodchip		3%	1.24bc	–15		
				Vineyard waste		3%	1.27cd	–13		
				Vineyard waste	400		1.23abc			
	Silt loam			Control		0%	1.36ef			
				Woodchip		3%	1.22abc	–11		
	Clay loam			Control		0%	1.31de	–4		
				Woodchip		3%	1.18ab	–15		

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Table 1. (cont.)

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Bulk density	Decrease in bulk density	Increase in porosity	Reference‡
					°C		Mg m^{-3}	%	%	
Poland	Loamy sand	Greenhouse	3 mo	*Miscanthus* × *giganteus* and winter wheat	300	0.5%	1.599a			Glab et al. (2016)
						1%	1.547b	–3	5	
						2%	1.466c	–9	12	
						4%	1.330d	–20	24	
USA	Loam, clay loam, and silty clay loam	Field	3 yr	Hardwood	~400	0, 9.9, and 18.4 Mg ha^{-1}	No effect		n/a	Rogovska et al. (2016)
China	Loam	Field	4 yr	Peanut shells	350–500	0 Mg ha^{-1}	1.36a	–4	15	Du et al. (2016)
						28 Mg ha^{-1}	1.31b			
Japan	Loam	Pots	100 d	Rice husk		0%	1.29a		Increased	Pratiwi and Shinogi (2016)
					750	2%	1.25a	–12		
						4%	1.13b			
China	Clay loam	Field	3 yr	Corn straw and peanut hulls	n/a	0 Mg ha^{-1}	1.36a	–10	n/a	Ma et al. (2016)
						7.8 Mg ha^{-1}	1.24b			
China	Clay loam	Field	2 yr	Corn residue	400	0 Mg ha^{-1}	1.35a			Xiao et al. (2016)
						10 Mg ha^{-1}	1.30ab			
						20 Mg ha^{-1}	1.24b	–9		
						30 Mg ha^{-1}	1.23b	–10	13	
Italy	Silty clay loam	Field	3 mo	Wheat bran		0 Mg ha^{-1}	1.07a			Andrenelli et al. (2016)
					800	14 Mg ha^{-1}	0.93b	–15		
					1200		0.96b	–11	7	

† Mesquite, *Prosopis juliflora* (Sw.) DC. rice, *Oryza sativa;* wheat, *Triticum aestivum;* corn, *Zea mays;* cotton, *Gossypium hirsutum* L.; red grama, *Bouteloua trifida* Thurb.; peanut, *Arachis hypogaea* L.

‡ Only studies published in 2016 are summarized in this table. Studies published prior to 2016 were reported by Omondi et al. (2016).

§ The lowercase letters for mean bulk density values indicate significant differences.

¶ n/a, not available.

soil by 2 to 41%. As expected, the decrease in soil bulk density with biochar application directly resulted in an increase in soil porosity. Soil porosity increased linearly with an increase in biochar application (Table 1). The findings from a synthesis of recent studies agree with the review by Omondi et al. (2016), which found that biochar addition increased soil porosity by 8.4%. Similar to bulk density, biochar appears to increase porosity more in coarse-textured soils than in fine-textured soils.

Biochar can increase soil porosity by (i) reducing soil bulk density, (ii) increasing soil aggregation, (iii) interacting with mineral soil particles, and (iv) reducing soil packing. Biochar particles have a porosity of 70 to 90%. Addition of this porous material to soil can concomitantly increase soil porosity. Some studies have reported that the porosity of biochar may be a function of pyrolysis temperature and could increase with an increase in temperature. For example, Andrenelli et al. (2016) reported that soil porosity was unaffected with the application of wheat (*Triticum aestivum*) bran pyrolized at 800°C but increased with the addition of wheat bran pyrolized at 1200°C.

The literature review herein indicates that biochar application generally reduces bulk density and particle density and increases soil porosity. The significant and positive effects of biochar on these properties can be primarily attributed to the low density of biochar particles. The effects of biochar on soil density and porosity appear to occur regardless of biochar type, the duration of the study, and soil type, but sandy soils appear to be more responsive to biochar application than soils with high clay content. The increased soil porosity caused by biochar application can have positive implications for the movement of water, heat, and gases in the soil. However, it is also important to consider that the lower bulk density and higher total porosity may not be always beneficial for plant growth. Though excessive soil compaction reduces aeration, water movement, plant root growth, and crop yields, slight compaction can improve root–soil contact and pore connectivity, allowing higher nutrient and water transport and supply to plant roots (Arvidsson, 1999).

Penetration Resistance

Penetration resistance is a dynamic indicator of soil strength or compaction and simulates the soil's resistance to penetration by plant roots. The review of published studies in Table 2 indicates the following. Field-scale studies of biochar and soil penetration resistance are not available. The few studies were conducted in the greenhouse and small plots. The available studies are short-term (<2 yr). Most studies were conducted for a few months. Such short-term studies

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may not provide a full understanding of soil–biochar interactive effects on soil properties. Biochar may require long periods of time to react with soil inorganic particles and make changes in soil compactibility. Biochar addition may not generally reduce soil penetration resistance. Five of the six studies in Table 1 found no significant effect of biochar addition on penetration resistance, although one found a decrease in penetration resistance. Busscher et al. (2010) found that penetration resistance decreased with biochar application at a rate of 44 Mg ha^{-1} but not at lower rates of application. This finding strongly suggests that large rates of biochar application may be necessary to significantly change the penetration resistance of the soil and thus reduce the risks of soil compaction. However, it should be noted that high rates of biochar application may not be economically viable. Soil textural class appears to have little or no influence on biochar effects on penetration resistance, but data are too limited to make definitive conclusions. The few data indicate that biochar has small or no effects on penetration resistance.

Tensile Strength

Tensile strength is a parameter of the soil strength and refers to the inherent ability of soil to resist the disruptive forces that cause fracture or rupture of the soil. Changes in tensile strength, similar to penetration resistance, can influence soil tillability, seedling emergence, root growth, and other soil processes. Because compacted, cemented, clayey, and low organic matter soils often have high values of tensile strength, addition of C-enriched materials such as biochar could reduce tensile strength. Table 2 shows that, similar to penetration resistance, soil tensile strength has been little studied under biochar application. The data summarized in Table 2 indicate the following.

- Biochar application can, in general, reduce tensile strength by 42 to 242% regardless of soil textural class.
- Biochar can particularly reduce the tensile strength of soil when applied at high rates: >2% (Zong et al., 2016) or >50 Mg ha^{-1} (Chan et al., 2007) of biochar, indicating that low rates of biochar may have limited or no effects on reducing soil strength.
- In some sandy soils, application of biochar and animal manure together can improve soil structure by bonding soil particles into strong aggregates and thus increasing tensile strength (Khademalrasoul et al., 2014). Organic amendments have aggregating effects in soils with low aggregate strength. Biochar could positively interact with other organic amendments (e.g., animal manure or compost), but the interactive mechanisms deserve further assessment.

Changes in tensile strength strongly depend on soil porosity, interparticle bonds, internal friction, forces, clay content and mineralogy, cementing agents, microstructural characteristics, and soil organic C. Thus the decreased tensile strength of soil following biochar addition indicates that biochar probably weakens the interparticle bonding and reduces the density and overall co-

Table 2. Impacts of biochar application on penetration resistance and tensile strength of soil.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Biochar effect	Reference
					°C			
					Penetration resistance			
Canada	Clay, loam, and sand	Greenhouse	113 d	Sugar maple and wild cherry	800–900	50.4–71.9 g kg^{-1}	No effect	Bekele et al. (2015)
USA	Loam	Small plots	2 yr	Mixed hardwood	500–575	0 and 96 Mg ha^{-1}	No effect	Rogovska et al. (2014)
USA	Silty clay loam	Small plots	1.5 yr	Oak	650	0.5%	No effect	Mukherjee et al. (2014)
USA	Silt loam	Small plots	1 yr	Oak	n/a‡	0, 5, or 25 Mg ha^{-1}	No effect	Eastman (2011)
USA	Sandy loam	Incubation	94 d	Pecan	700	0 and 22 Mg ha^{-1}	No effect	Busscher et al. (2011)
USA	Loamy sand	Incubation	96 d	Pecan	700	0, 11, 22, and 44 Mg ha^{-1}	Reduced at 44 Mg ha^{-1}	Busscher et al. (2010)
					Tensile strength			
China	Clay loam	Incubation	180 d	Straw, woodchips, and wastewater sludge	500	2, 4, and 6%	Reduced by 100–184% under 6% biochar addition	Zong et al. (2016)
China	Clayey soil				500		Reduced by 42–100% under 6% biochar addition	Zong et al. (2014)
China	Clayey soil	Incubation	180 d	Rice husk biochar	n/a	2, 4, and 6%	Reduced by 164%	Lu et al. (2014)
Denmark	Sandy loam	Small plots	1.5 yr	Birch	500	0 and 100 Mg ha^{-1} + swine manure	Increased on 4–16 mm aggregates	Khademalrasoul et al. (2014)
Australia	Alfisol	Greenhouse	6 wk	Grass clippings, cotton trash, and plant prunings		0 Mg ha^{-1}	64.4a	Chan et al. (2007)
						10 Mg ha^{-1}	69.2a	
					450	50 Mg ha^{-1}	31.7b (103% decrease)	
						100 Mg ha^{-1}	18.8c (242% decrease)	

† Sugar maple, *Acer saccharum* Marshall; wild cherry, *Prunus avium* (L.) L.; oak, *Quercus* sp.; pecan, *Carya illinoinensis* (Wangenh.) K. Koch; rice, Oryza sativa; birch, *Betula* sp.; cotton, *Gossypium hirsutum*.

‡ n/a = not available.

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hesiveness of the soil (Chan et al., 2007; Busscher et al., 2011; Zong et al., 2016). Thus addition of biochar can contribute to microstructural development and soil resilience against external forces (Ajayi and Horn, 2016).

The general reduction in tensile strength and the little or no difference in penetration resistance with biochar application (Table 2) appear to suggest that soil tensile strength can be more sensitive to biochar application than soil penetration resistance. It is well recognized that penetration resistance is highly variable on temporal and spatial scales and strongly depends on soil water content at the time of measurement, requiring adjustments for differences in soil water content among study treatments prior to data interpretation. In addition, penetration resistance is commonly measured under in situ conditions on a large volume of soil using penetrometers, whereas tensile strength is often measured in the laboratory using intact soil cores, molded soil samples, or individual soil aggregates. Tensile strength could be less susceptible to variability and make a better measure of soil strength than penetration resistance. Overall, the tensile strength of the soil decreases with biochar application and this could be important for ease of tillage operations, seedling emergence, and root growth.

Shear Strength

Shear strength, which refers to the internal resistance of soil against sliding as a function of cohesion and the angle of internal friction, is another important soil mechanical property. Unlike other soil physical properties, changes in soil shear strength following biochar application have not been studied much. The few published studies suggest that biochar effects could vary with soil and feedstock type. Biochar application appears to consistently reduce shear strength in soils with a high clay content. In an incubation study using a clayey soil, Zong et al. (2014) found that application of three types of biochar including woodchip, straw, and wastewater sludge at 0, 2, 4, and 6% reduced shear strength by reducing cohesion and the internal friction angle. In a similar incubation study using a clay loam soil, Zong et al. (2016) found that woodchip biochar application at 4 and 6% reduced soil cohesion compared with 0 and 2% application, thereby reducing shear strength; but application of other biochars such as straw and wastewater sludge biochar had no effect. In a laboratory study, Reddy et al. (2015) reported that wood biochar had higher shear strength than a silty clay soil alone. Thus mixing biochar at rates of 5, 10, and 20% with the silty clay soil resulted in an increase in the shear strength of the mixture. Field studies are needed to corroborate the findings from the laboratory or greenhouse experiments that indicate some positive effects of biochar on the shear strength of clayey soils.

Soil Consistency (Atterberg Limits)

The consistency limits or Atterberg limits are important indicators of the mechanical and hydrological behavior of soils. The liquid limit, plastic limit, and plasticity index are measures of soil consistency. Information on soil consistency (Atterberg limits) is essential to managing soils for engineering (e.g., the stability of building foundations) and agronomic (e.g., compaction and tillage operations) purposes. Similar to other soil physical properties, Atterberg limits depend on changes in soil organic C concentrations (Blanco-Canqui et al., 2006). As a result, the application of biochar to soil can directly influence the Atterberg limits by increasing soil C concentration. Although published data on biochar and rheological properties are few and primarily come from incubated experiments, they provide some insights into biochar effects on rheological properties. In a sandy soil, application of three biochars made from straw, woodchips, and wastewater sludge at 0, 2, 4, and 6% application rates significantly increased the liquid limit by 8 to 22% and the plasticity index by 48 to 99%, but the plastic limit tended to decrease, although no significant effects were seen (Zong et al., 2016). The same study found that straw biochar was more effective at increasing the liquid limit and plasticity index than woodchip biochar and wastewater sludge biochar, suggesting that biochar effects on soil consistency can depend on the type of biochar feedstock. The plasticity index, which is the difference between plastic and liquid limits, is an indicator of the range of water contents at which a soil can be susceptible to compaction (Zong et al., 2016). Likewise, on a clayey soil, application of rice husk biochar at 0, 2, 4, and 6% increased the plastic and liquid limits by 15 to 18% but had no effect on the plasticity index (Lu et al., 2014). The review suggests that application of biochar can generally improve soil consistency regardless of soil textural class, but the extent of effects can depend on biochar feedstock type.

Coefficient of Linear Extensibility and Cracking

Soils with a high clay content can have a high shrink–swell potential, which can cause soil cracking. Although cracking can be beneficial for rapid water infiltration and groundwater recharge in clayey soils, it can cause loss of water and nutrients from the root zone, thereby reducing soil productivity. How does biochar application affect the shrinkage, swelling, and cracking of soils? The few studies on this topic indicate that biochar can ameliorate soil shrinkage, swelling, and cracking. In a greenhouse experiment, Zong et al. (2014) found that wheat straw, woodchip, and wastewater sludge biochar applied at rates of 0, 2, 4, and 6% reduced the coefficient of linear extensibility (COLE) of a clayey soil. The same study found that biochar reduced soil cracking, and the surface area and length of the cracks. Similarly, Lu et al. (2014) reported that rice husk biochar applied at 0, 2, 4, and 6% decreased the COLE of a clayey soil in an incubation study. Application of biochar at 6% reduced the COLE from 0.63 to 0.56. These findings indicate that biochar can be a potential amendment to improve physical quality of soils with expandable clays. Similar to other soil mechanical and compaction parameters, field data are needed to better evaluate the extent of biochar effects on soils with a high shrink–swell potential.

BIOCHAR AND SOIL STRUCTURAL PROPERTIES

Wet Aggregate Stability

Wet aggregate stability is a sensitive indicator of soil structural stability and resilience. It affects many soil processes, including development of macropores, water infiltration, water erosion, and others. Table 3 summarizes the findings of studies on wet aggregate stabil-

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ity expressed as the mean weight diameter of aggregates and the percentage of water-stable aggregates. The published studies on both parameters were conducted across 34 soils covering a wide range of textural classes from sand to clay. The findings show that biochar application increased wet aggregate stability in 24 soils but had no effect on 10 soils (Table 3). Biochar application increased the mean weight diameter of water-stable aggregates by 4 to 58% and the proportion of water-stable aggregates by 21 to 226% (Table 1). The rate of biochar used in the studies ranged from 0.1 to 10% and from 5 to 96 Mg ha^{-1}. The improvement in wet aggregate stability in the 26 soils generally increased as the amount of biochar increased.

According to the data in Table 3, the lack of significant effects of biochar application in the 10 studies is not related to the amount of biochar applied. Application of biochar at rates as high as 96 Mg ha^{-1} in the US Midwest (Rogovska et al., 2014) and 5% (about 140 Mg ha^{-1}) in Denmark (Hansen et al. (2016) did not result in increased wet aggregate stability. Although the summary in Table 3 showed no consistent trend in changes in wet aggregate stability by textural class, Burrell et al. (2016) and Ouyang et al. (2013) have indicated that biochar can increase wet aggregate stability more in sandy than in silty clay or clayey soils. The organic particles of biochar may improve the particle bonding of large particles and promote soil aggregation in coarse-textured rather than in fine-textured soils. It is generally considered that biochar could have greater positive effects on C-depleted soils than on soils with a high organic C concentration, but a better understanding of this relationship is needed. The initial soil organic C level may not be the only factor that affects changes in wet aggregate stability after biochar addition. In some cases, soils with organic C concentrations as low as 0.43% may not exhibit any changes in wet aggregate stability after biochar application (Dong et al., 2016). On the other hand, soils with organic C concentrations as high as 10% on a grassland in New Zealand (Herath et al. (2013) and 4.5% on a volcanic soil in Chile (Curaqueo et al., 2014) exhibited greater soil aggregate stability than the controls after biochar addition.

The review suggests that biochar application can, in general, increase wet aggregate stability regardless of the differences in soil textural class and initial soil organic C concentration. The lack of significant changes in wet aggregate stability observed in some studies suggests that biochar effects can depend on site-specific characteristics. Interactions among texture, soil slope, organic C, biochar properties, climate, and others could dictate the extent to which biochar application can change soil aggregate stability. For example, in sloping soils, biochar may have a limited effect on improving soil aggregate stability, as surface-applied biochar can be lost during water erosion, particularly when the biochar is not completely incorporated into the soil. Similarly, in flat and wind-erosion prone soils, biochar can be lost through wind erosion (Peng et al., 2016). Differences in clay mineralogy as well as climate could be fac-

Table 3. Biochar impacts on wet soil aggregate stability for different scenarios of soil and biochar management in field and laboratory experiments.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyr1olysis temperature	Biochar rate	Aggregate stability	Change	Reference
					°C			%	
					Mean weight diameter of aggregates, mm				
Sri Lanka	Sand	Laboratory	6 mo	Rice husk		0, 0.1, and 0.5%	~0.70b‡		Gamage et al. (2016)
						1%	~0.83a	19	
	Sandy loam				~600	0%	1.03b		
						0.1%	1.20c	16	
						0.5%	1.34b	30	
						1%	1.60a	55	
Iran	Sandy loam	Incubation	180 d	Rice husk and wood	350–550	0 and 2%	Increased		Esmaeelnejad et al. (2016)
China	Sandy loam Silty clay (1.2)	Incubation	≤90 d	Dairy manure	500	0 and 2%	Increased		Ouyang et al. (2013)
New Zealand	Silt loam	Greenhouse	295 d	Corn stover	350–550	10–17 Mg ha^{-1}	Increased	4–17	Herath et al. (2013)
Chile	Silt loam	Field	195 d	Oat hull		0 Mg ha^{-1}	2.7b		Curaqueo et al. (2014)
					700	5 Mg ha^{-1}	3.05ab	13	
						10 Mg ha^{-1}	3.17ab	17	
						20 Mg ha^{-1}	4.16a	54	
	Loamy silty clay					0 Mg ha^{-1}	2.87c		
						5 Mg ha^{-1}	3.15c	10	
						10 Mg ha^{-1}	3.99b	39	
						20 Mg ha^{-1}	4.33a	51	

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Table 3. (cont.)

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyr1olysis temperature	Biochar rate	Aggregate stability	Change	Reference
					°C			%	
China	Clay loam	Field	3 yr	Corn straw and peanut hulls	n/a¶	0 Mg ha^{-1}	1.30b		Ma et al. (2016)
						8 Mg ha^{-1}	1.75a	35	
China	Clay	Greenhouse	180 d	Rice husk		0%	0.92c§		Lu et al. (2014)
					n/a	2%	0.91d	–1	
						4%	1.07b	16	
						6%	1.14a	24	
China	Clay	Greenhouse	180 d	Straw		0%	0.47bc		Sun and Lu (2014)
						2%	0.41c		
						4%	0.51ab	8	
						6%	0.57a	21	
				Woodchips		0%	0.47a		
					500	2%	0.41b	–15	
						4%	0.44ab		
						6%	0.46a		
				Wastewater–sludge		0%	0.47b		
						2%	0.48b		
						4%	0.47b		
						6%	0.60a	28	
China	n/a	Plot	4 yr	n/a	n/a	67 Mg ha^{-1}	Increased	15	Cui et al. (2016)
China	n/a	Field	1 yr	Wheat straw	350–550	0–40 Mg ha^{-1}	Increased	9–38	Liu et al. (2014)
China	n/a	Pot	7 mo	Corn straw		0%	Increased		Hua et al. (2014)
					600	3%		8–20	
						8%		40–58	
					Amount of water- stable aggregates, %				
China	Silt loam	Incubation	20 d	Sugarcane bagasse		0%	8.6e		Abdelhafez et al. (2014)
						1%	14cd	63	
						2.5%	19b	121	
						5%	24a	179	
					500	10%	28a	226	
				Orange peel		1%	11de	28	
						2.5%	13d	51	
						5%	16bc	86	
						10%	19b	121	
China	Clay loam	Incubation	372 d	Wood		0%	44.3bc		Demisie et al. (2014)
						0.5	48.2b		
						1.0	43.0bc		
					600	2.0	39.8c		
				Bamboo		0.5	55.4a	25	
						1.0	53.6a	21	
						2.0	40.3c		
Finland	Silty clay, clay	Incubation	3 wk	Wood	550–600	0–30 Mg ha^{-1}	Increased		Soinne et al. (2014)

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Table 3. (cont.)

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Aggregate stability	Change	Reference
					°C			%	
South Korea	Silt loam	Pot	<7 wk	Rice hull	500	0–5%	Increased	30–67	Kim et al. (2016)
Zambia	Sandy loam, loamy sand, sand	Field	1 yr	Corn cob and rice husk	350	0–4%	Increased	28–80	Obia et al. (2016)
Italy	Clay	Laboratory	n/a	Wood	1200	42–85 Mg ha^{-1}	Increased		Baiamonte et al. (2015)
Austria	Sandy loam	Greenhouse	3 yr	Control		0 and 3%	Increased		Burrell et al. (2016)
				Straw				92	
				Wood				50	
				Vineyard waste	400			37	
				Vineyard waste	525			28	
	Silt loam			Control					
				Wood				26	
				Mean weight diameter of aggregates					
China	Sandy loam	Field	1 yr	Corncob	450–550	4.5 and 9 Mg ha^{-1}	No effect		Zhang et al. (2015)
China	Silt loam	Field	3 yr	Rice husks and cotton seed hulls	400	0 to 89 Mg ha^{-1}	No effect		Dong et al. (2016)
USA	Loam	Field	2 yr	Hardwood	500–575	0 and 96 Mg ha^{-1}	No effect		Rogovska et al. (2014)
Germany	Sandy or silty soil	Pot or field	826 d	Hard wood	550	0 and 1.5%	No effect		Borchard et al. (2014)
China	Clay loam	Field	<3 yr	Rice straw	400	21 Mg ha^{-1}	No effect		Peng et al. (2016)
China	Clay loam	Pot	11 d	Rice straw	250–450	0 and 1%	No effect		Peng et al. (2011)
				Amount of water-stable aggregates					
The Netherlands	Sand	Field	5 mo	Hay	400–600	0–50 Mg ha^{-1}	No effect		Jeffery et al. (2015)
Denmark	Sandy loam	Incubation	2 yr	Wood and wheat straw	750–1200	0 and 5%	No effect		Hansen et al. (2016)
Austria	Clay loam	Greenhouse	3 yr	Woodchip	400–525	0 and 3%	No effect		Burrell et al. (2016)

† Rice, *Oryza sativa*; corn, *Zea mays*; oats, *Avena sativa* L.; peanut, *Arachis hypogaea*; sugarcane, *Saccharum officinarum* L.; wheat, *Triticum aestivum*; bamboo, *Bambusa* sp.; orange, *Citrus* sp.

‡ The lowercase letters for mean aggregate stability values indicate significant differences.

§ Slow wetting of 2–6 mm aggregates.

¶ n/a = not available.

tors influencing the response of soil aggregate stability to biochar application.

The increase in wet aggregate stability with biochar observed in most soils suggests that biochar could enhance soil aggregation by providing organic binding agents. It is well understood that soil aggregate stability is strongly correlated with organic matter concentration. Carbon particles can form organic ligands to the surface of soil particles in combination with bonding of polyvalent cations. The coalescence of organic and inorganic particles (organo-mineral complexes) results in the formation of soil microaggregates and then macroaggregates. Clay particles may bond more strongly with organic particles to form aggregates because of the greater specific surface area of clay particles compared with sand particles. Biochar particles are the organic core around which the inorganic particles can bind and form soil aggregates through physical, chemical, and biological forces (Hua et al., 2014). Overall, biochar appears to have the ability to enhance soil aggregation and improve the soil's structural quality.

Dry Aggregate Stability

Dry soil aggregate stability is another important indicator of soil structural development. It is a property that influences soil erodibility against wind erosion, crusting, and root growth, among others. For example, soils with small dry soil aggregates have a large proportion of the wind-erodible fraction (<0.84 mm diam. aggregates) and are thus more susceptible to wind erosion than those with large and stable dry aggregates. The literature review indicates that biochar application has mixed effects on dry aggregate stability (Table 4). Biochar application increased dry aggregate stability in three soils but had no effect on five soils of the eight soils summarized in Table 4. It appears that biochar has less effect on increasing dry aggregate stability in

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sandy soils than in sandy loams or silt loams. For example, when comparing four soil textural classes, Liu et al. (2012) reported that biochar did not increase dry aggregate stability in soils with a sand content greater than 17.3% but it increased dry aggregate stability in a soil with 7.9% sand content.

Sandy soils may require large applications of biochar before changes in dry aggregate stability are measurable relative to sandy loams or medium-textured soils. Gamage et al. (2016) found that dry aggregate stability in a sandy loam increased linearly with biochar application rates below 1%, but in a sandy soil, dry aggregate stability increased only when 1% biochar (about 25 Mg ha^{-1}) was applied. The linear increase in dry aggregate stability with biochar addition in the sandy loam suggests that biochar may bond or interact with particles even at low rates relative to sandy soils. The minimum threshold level of application at which biochar improves soil properties and the maximum threshold level at which biochar no longer improves soil properties should be determined for each soil. Liu et al. (2012) reported that 0.8% of biochar increased dry aggregate stability relative to 0% application but 1% biochar application had no effect, suggesting that large applications may not always improve dry aggregate stability in some soils.

Initial soil organic C concentrations appear to have no clear influence on biochar benefits. According to the studies by Liu et al. (2012) and Gamage et al. (2016), dry aggregate stability following biochar application tends to increase more in soils with a higher rather than a lower initial soil C concentration. The available studies suggest that biochar application may have positive or no effects on improving dry aggregate stability. Biochar particles could improve dry soil aggregation in the long term, as biochar particles oxidize and react with soil particles, but this hypothesis necessitates further long-term experimentation under different soils and biochar application rates.

BIOCHAR AND SOIL HYDRAULIC PROPERTIES

Water Repellency

Soil water repellency directly influences hydrological processes including runoff, bypass flow, water infiltration, water retention, macropore integrity, and others. It does not only alter water entry and distribution in the soil but also influences other processes including aggregation, microbial activity, and organic matter decomposition. The latter processes, in turn, influence soil physical processes. Although severe soil water repellency can reduce infiltration and adversely affect related hydrological processes (Briggs et al., 2012), subcritical or slight water repellency such as that observed in agricultural soils (e.g., no-till management) is important for water uptake, soil aggregate stabilization, soil C protection, and macropore preservation. Rapid water entry into hydrophilic soil causes both air entrapment and slaking of soil aggregates, which can concurrently clog the pores and reduce water infiltration.

Forest soils affected by fires can be highly hydrophobic, which has been widely studied (DeBano, 2000; Briggs et al., 2012). The question is how biochar application affects soil water repellency. The biochar microstructure is often highly hydrophobic (Kinney et al., 2012), which suggests that when it is added to soil, biochar can induce water repellency. The few data available to this point suggest that biochar has little or no effect on the water repellency of soils (Table 5). Three out of the five studies in Table 5 found no effects of biochar on water repellency. The two remaining studies found contrasting effects (Devereux et al., 2012; Głąb et al., 2016). Devereux et al. (2012) reported that biochar application at 1.5, 2.5, and 5.0% reduced water repellency and the 5.0% rate had the largest effect, reducing water repellency fivefold compared with the control. However, Głąb et al. (2016) found that biochar application at 4.0% slightly increased

Table 4. Impacts of biochar application on dry soil aggregate stability for different soils.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Dry aggregate stability	Reference
					°C		%	
USA	Sandy loam	Incubation	94 d	Pecan	700	0 and 22 Mg ha^{-1}	No effect	Busscher et al. (2011)
China	n/a§	Plot	4 yr	n/a	n/a	67 Mg ha^{-1}	No effect	Cui et al. (2016)
China	Sandy loam	Incubation	11 mo	Wood		0, 0.4, 0.8, and 1.6%	No effect	Liu et al. (2012)
	Sandy loam						No effect	
	Silt loam						No effect	
	Silt loam				660	0%	30.2b‡	
						0.4%	33.9ab	
						0.8%	44.0a	
						1.6%	34.0ab	
Sri Lanka	Sand	Laboratory	6 mo	Rice husk		0	Increased with 1% only	Gamage et al. (2016)
						0.1%		
						0.5%		
					~600	1%		
	Sandy loam					0	Increased at all biochar rates	
						0.1%		
						0.5%		
						1%		

† Pecan, *Carya illinoinensis*; rice, Oryza sativa.

‡ The size of dry soil aggregates is >10 mm.

§ n/a, not available.

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water repellency compared with 0.5% application but 1 and 2% application had no effect. The water repellency of soils can be classified as: wettable [water drop penetration test (WDPT) < 5 s], slightly repellent (WDPT = 5–60 s), strongly repellent (WDPT = 60–600 s), severely repellent (WDPT = 600–3600 s), and extremely repellent (WDPT > 3600 s; Dekker and Jungerius, 1990). The study by Devereux et al. (2012) observed that biochar reduced the WDPT of the soil from 11 s to 4 and 2 s, indicating that biochar application reduced soil from slightly water repellent to non-water-repellent. The study by Głąb et al. (2016) observed that biochar increased the WDPT of the soil from 1.02 to 1.62 s, but this increase was very small and the soil–biochar mixture is classified as non-water-repellent and wettable.

Interactions of biochar production temperature, biochar placement method, soil texture, soil water content, time after application, and others may influence biochar effects on water repellency. In general, fresh biochar can be more water repellent than old biochar (Briggs et al., 2012). Additionally, pyrolysis temperature can influence biochar hydrophobicity. Biochar produced at low temperatures can be more water repellent than those produced at high temperatures. Kinney et al. (2012) found that corn (*Zea mays* L.) stover, magnolia (*Magnolia grandiflora* L.) leaf, and apple (*Malus* sp.) wood biochar produced at 300°C was 13 times more hydrophobic than biochar produced at 500°C. High pyrolysis temperatures reduce hydrophobicity by removing organic compounds (e.g., aliphatic functional groups) from the surface of biochar particles (Gray et al., 2014). The extent of changes in repellency with an increase in pyrolysis temperature can also depend on biochar feedstock. For example, Kinney et al. (2012) observed that the decrease in water repellency with increasing pyrolysis temperature was larger for biochar from corn stover and apple wood than biochar from magnolia leaf. Page-Dumroese et al. (2015) observed that biochar mixed with the soil induced less water repellency than surface-applied biochar in both coarse-textured and fine-textured soils, particularly when soil water content increased above 25%. The severity of water repellency could decrease with time after application, as biochar–soil contacts improve and water-repellent layers degrade. In summary, the review indicates that biochar application can have little or no effect on inducing soil water repellency.

Water Infiltration, and Saturated and Unsaturated Hydraulic Conductivity

Increasing the rate at which water enters the soil is important to precipitation capture, water storage, and overall soil water management. The literature review shows that fewer published data are available on water infiltration (Table 6) and unsaturated hydraulic conductivity than on saturated hydraulic conductivity (Table 7). Studies indicate that biochar application can alter water infiltration, saturated hydraulic conductivity, and unsaturated hydraulic conductivity. Table 7 shows that out of 28 study soils, biochar altered the saturated hydraulic conductivity in 22 soils. Out of 15 coarse-textured soils (coarse sand, sand, fine sand, and sandy loam soils), biochar reduced the saturated hydraulic conductivity in 12 of them. The rate of decrease in saturated hydraulic conductivity ranged from 7 to 2270%. This wide range in saturated hydraulic conductivity decrease is not surprising as this soil property has a high coefficient of variation. The mean decrease in saturated hydraulic conductivity across the 12 coarse-textured soils was 392%. The few studies on unsaturated hydraulic conductivity found similar results to those on saturated hydraulic conductivity. Understanding how biochar application affects water flow when the soil is unsaturated is important, considering that soils remain unsaturated most of the time. In a sandy soil, Uzoma et al. (2011) found that wood biochar application reduced unsaturated hydraulic conductivity with increasing soil water content and suggested that addition of biochar at 20 Mg ha^{-1} can be more effective than at a rate of 10 Mg ha^{-1}.

Table 7 also indicates that biochar increased the saturated hydraulic conductivity in 8 out of 13 fine-textured soils (loam, silt loam, silty clay loam, clay loam, and clay). The increase in saturated hydraulic conductivity in these soils ranged from 25 to 328% with a mean increase of 98%. This mean increase (98%) under fine-textured soils was lower than the mean decrease (392%) under coarse-textured soils, which strongly suggests that biochar has greater effects on saturated hydraulic conductivity in coarse-textured soils than in fine-textured soils. In clayey soils, Kameyama et al. (2012)

Table 5. Biochar impacts on soil water repellency under different management conditions.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Water repellency	Reference
					°C			
Germany	Sand	Lab and field	<6 mo	Corn	750	1, 2.5, and 5%	No effect	Abel et al. (2013)
Italy	Sandy clay loam	Plots	2 yr	Orchard prunings	500	22 and 44 Mg ha^{-1}	No effect	Baronti et al. (2014)
United Kingdom	Sandy loam	Greenhouse	<10 wk	Wood		0%	11 s a‡	Devereux et al. (2012)
					n/a	1.5 and 2.5%	4.1 s b	
						5%	2 s c	
Poland	Loamy sand	Greenhouse	3 mo	*Miscanthus* × *giganteus* and winter wheat		0.5%	1.02 s	Głąb et al. (2016)
					300	1%	1.03 s	
						2%	1.15 s	
						4%	1.62 s	
New Zealand	Silt loam	Greenhouse	295 d	Corn stover at	300–550	17.3, 11.3 and 10.0 Mg ha^{-1}	No effect	Herath et al. (2013)

† Corn, *Zea mays*; wheat, *Triticum aestivum*.

‡ The lowercase letters next to the mean values indicate significant differences.

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Table 6. Studies reporting the impacts of biochar application on water infiltration for different soils.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Infiltration	Reference
					°C			
Saudi Arabia	Sandy loam	Pots	5 wk	Wood		0%	0.763a‡§	Ibrahim et al. (2013)
						0.5%	0.761a	
					400	1.0%	0.548b	
						1.5%	0.564c	
						2.0%	0.534d	
USA	Sandy loam	Greenhouse	<2 mo	Peanut hulls	500	0, 25, 50, 75, and 100% by volume	Gradually decreased	Githinji et al. (2014)
USA	Compacted sandy loam	Incubation	128 d	Wood		0%	0.095 mL min^{-1}	Novak et al. (2016)
					500	2.0%	0.16–0.22 mL min^{-1}	
Australia	Clay loam	Plots	2 yr	Tree residues	600	0 and 20 Mg ha^{-1}	Increased	Prober et al. (2014)
USA	Loamy sand	Incubation	96 d	Pecan	700	0, 11, 22, and 44 Mg ha^{-1}	No effect	Busscher et al. 2010
USA	Loam	Plots	2 yr	Wood	500–575	0 and 96 Mg ha^{-1}	No consistent effect	Rogovska et al. (2014)

† Peanut, *Arachis hypogaea*; pecan, *Carya illinoinensis*.

‡ Sorptivity or initial infiltration (cm $min^{-0.5}$)

§ The lowercase letters next to the mean values indicate significant differences.

reported that biochar application at 5 and 10% by weight increased unsaturated hydraulic conductivity at all matric potentials.

The data on water infiltration and hydraulic conductivity suggest three trends. Addition of biochar:

- reduces water infiltration and hydraulic conductivity in sandy loams;
- increases water infiltration and hydraulic conductivity in clay loam or compacted soils; and
- has limited or no effects on medium-textured soils.

The first trend suggests that mixing biochar particles with coarse-textured soils could reduce water infiltration and hydraulic conductivity. This could be caused by the filling or clogging of soil macropores with fine biochar particles. Biochar material applied is often <2 mm in diameter. The small particles can fill the pore space and interact with soil inorganic particles. Biochar particles could alter the porous media of the soil, which directly influence the rate of water flow. In addition, some unstable biochar particles may rapidly disintegrate, cement, and clog the soil pores, reducing water flow within the soil. In particular, plate-like biochar particles may readily clog soil micropores than large and spherical particles (Githinji, 2014; Novak et al., 2016). Biochar with hydrophobic properties can also induce water repellency and reduce infiltration and hydraulic conductivity. On a sandy soil, Zhang et al. (2016) found that ground biochar (powder) reduced saturated flow more than biochar with large particles (5–8 mm diam.) under the same biochar rate. Biochar could also reduce water flow in sandy soils by improving bonding of sand particles, increasing soil cohesiveness, and adsorbing water. Sandy soils can be more responsive to biochar than clayey soils because of the larger differences in particle size between biochar and sand particles. Biochar application rates as low as 0.5% appeared to reduce saturated flow in sandy soils, but only >1% application rates appeared to increase saturated hydraulic conductivity in soils with high clay content (Table 7).

The second trend suggests that application of biochar to fine-textured or compacted soils can improve water flow. Mixing of large biochar particles (<2 mm) with predominantly smaller soil inorganic particles (e.g., clay) can increase pore space and water flow (Githinji, 2014). Novak et al. (2016) found that biochar increased initial water infiltration in compacted soils. Porous biochar particles could improve water flow more than biochar particles with predominance of micropores. On fine-textured soils, biochar particle can also increase water infiltration and hydraulic conductivity by improving soil aggregation and thus increasing macroporosity. Particularly, the addition of biochar with particles larger than clay particles can create heterogeneity in pore size distribution, promoting aggregation and macroporosity and thereby increasing saturated flow through the soil. Clayey soils commonly have lower water flow rates than sandy soils. Thus biochar could be an important amendment to improve water movement in clayey soils.

The third trend indicates that although biochar decreases infiltration and hydraulic conductivity in sandy soils and increases infiltration and hydraulic conductivity in clayey or compacted soils, it may not affect water flow in medium-textured soils. Table 7 indicates that biochar had no effect on saturated hydraulic conductivity in six soils including three loamy soils, two clayey soils, and one sandy soil. Long-term field studies are needed for a better understanding of these trends in different textural classes. The available studies are short-term (<2 yr), which suggests that the observed changes in water flow may be mainly caused by the mixing effect. The lack of any effect of biochar primarily on medium-textured soils such as loamy soils appears to suggest that intermediate soil particle sizes between sand and clay may be the least responsive to biochar addition.

Studies reporting a decrease in water flow suggest that the extent of water flow decrease is a function of the biochar application rate (Table 6). Both Ibrahim et al. (2013) and Githinji (2014) found a gradual decrease in water infiltration with an increase in the biochar application rate. For example, Githinji (2014) found that cumulative water infiltration was negatively and significantly correlated with biochar rates (r^2 = 0.952), decreasing linearly as the biochar amount increased. The same study observed that application of 100% biochar by volume reduced cumulative infiltration by about 13 times, which was attributed to the use of biochar with high hydrophobic properties. Low rates of biochar may not reduce wa-

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Table 7. Impacts of biochar application on saturated hydraulic conductivity (K_{sat}) across different soils.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	K_{sat} ‡	Change	Reference
					°C		cm h^{-1}	%	
Decrease in Saturated Hydraulic Conductivity									
China	Sand	Column	<30 d	Wood		0 g kg^{-1}	48.84a§		Zhang et al. (2016)
					350–550	7 g kg^{-1}	48.84a	0	
						15 g kg^{-1}	42.96b	–14	
						25 g kg^{-1}	42.54b	–15	
USA	Sand	Laboratory	2 d	Switchgrass		0% v/v	84.8		Brockhoff et al. (2010)
						5% v/v	55.9	–52	
						10% v/v	53	–60	
					500	15% v/v	29.2	–190	
						20% v/v	15.5	–447	
						25% v/v	6.6	–1185	
Japan	Sand	Greenhouse	85 d	Cow manure		0 Mg ha^{-1}	7056a		Uzoma et al. (2011)
					300–500	10 Mg ha^{-1}	6588b	–7	
						15 Mg ha^{-1}	5076c	–39	
						20 Mg ha^{-1}	4572d	–54	
USA	Sand	Laboratory		Wood	400	100 g kg^{-1}	Decreased	–92	Barnes et al. (2014)
Mongolia	Sand	Laboratory	<7 d	Peanut shell		0	1.2a		Dan et al. (2015)
					n/a#	50 g kg^{-1}	0.2b	–500	
						100 g kg^{-1}	0.1b	–1100	
						150 g kg^{-1}	0.1b	–1100	
USA	Coarse sand	Laboratory	n/a	Hardwood§,		0	24.89 (1.94)		Lim et al. (2016)
						10 g kg^{-1}	6.98 (0.51)	–256	
						20 g kg^{-1}	3.19 (0.42)	–680	
						50 g kg^{-1}	1.05 (0.09)	–2270	
					500				
	Fine sand					0 g kg^{-1}	10.77 0.98)		
						10 g kg^{-1}	6.91 (0.09)	–56	
						20 g kg^{-1}	5.58 (0.1)	–93	
						50 g kg^{-1}	1.54 (0.03)	–599	
Iran	Sandy loam	Incubation	180 d	Rice husk and wood	350–550	20 g kg^{-1}	Decreased		Esmaeelnejad et al. (2016)
UK	Sandy loam	Greenhouse	<10 wk	Wood	n/a	0, 15, 25, and 50 g kg^{-1}	Decreased	–110 at 50 g kg^{-1}	Devereux et al. (2012)
USA	Sandy loam	Greenhouse	<2 mo	Peanut hulls	500	0, 25, 50, 75, and 100% v/v	Decreased		Githinji et al. (2014)
Saudi Arabia	Sandy loam	Pots	5 wk	Wood		0 g kg^{-1}	4.9a		Ibrahim et al. (2013)
						5 g kg^{-1}	4.8a		
					400	10 g kg^{-1}	4.5b	–9	
						15 g kg^{-1}	4.3c	–14	
						20 g kg^{-1}	4.1d	–19	
Germany	Fine sand	Columns	200 d	Wood		0	35.2a		Ajayi and Horn (2016)
						20 g kg^{-1}	26.8a	–31	
					500–600	50 g kg^{-1}	15.6b	–126	
						60 g kg^{-1}	8.0c	–340	
Increase in Saturated Hydraulic Conductivity									
Sri Lanka	Sand	Laboratory	6 mo	Rice husk		0, 1, 5, and 10 g kg^{-1}	Increased	54–78% at > 5 g kg^{-1}	Gamage et al. (2016)
	Sandy loam				~600			82–148 at > 5 g kg^{-1}	
Germany	Loamy sand	Laboratory	7 d	Corn and wood	750	0 and 20 g kg^{-1}	Increased	n/a	Eibisch et al. (2015)
Mongolia	Silt loam	Laboratory	<7 d	Peanut shell		0	0.008d		Dan et al. (2015)
					n/a	50 g kg^{-1}	0.010c	25	
						100 g kg^{-1}	0.025b	212	
						150 g kg^{-1}	0.031a	287	

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Table 7. (cont.)

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	K_{sat} ‡	Change	Reference
					°C		cm h^{-1}	%	
New Zealand	Silt loam	Greenhouse	295 d			0 Mg ha^{-1}	10.08c		Herath et al. (2013)
				Corn stover	350	11.3 Mg ha^{-1}	13.32b	32	
				Corn stover	550	10.0 Mg ha^{-1}	24.12a	139	
Germany	Silty clay loam	Columns	200 d	Wood		0	0.095b		Ajayi and Horn (2016)
					500–600	20 g kg^{-1}	0.119b	25	
						50 g kg^{-1}	0.153b	61	
						60 g kg^{-1}	0.168a	77	
USA	Clay loam	Greenhouse	30 d	Wood	500	0 and 75 Mg ha^{-1}	increased	180	Chaganti and Crohn (2015)
Laos	Clay loam	Field	1 yr	Wood		0	0.59b		Asai et al. (2009)
					n/a	4 Mg ha^{-1}	0.89b	51	
						8 Mg ha^{-1}	0.77b	30	
						16 Mg ha^{-1}	1.63a	176	
USA	Clay	Laboratory	n/a	Hardwood¶		0 g kg^{-1}	1.03 (0.09)		Lim et al. (2016)
						10 g kg^{-1}	1.44 (0.03)	40	
					500	20 g kg^{-1}	1.85 (0.03)	80	
						50 g kg^{-1}	1.02 (0.22)		
USA	Clayey	Laboratory		Wood	400	0 and 100 g kg^{-1}	Increase	328	Barnes et al. (2014)
No Effect on Saturated Hydraulic Conductivity									
The Netherlands	Sand	Field	5 mo	Hay	400–600	0, 1, 5, 20, and 50 Mg ha^{-1}	No effect		Jeffery et al. (2015)
USA	Loam	Laboratory	500 d	Wood	700	0, 5, 10, and 20 g kg^{-1}	No effect		Laird et al. (2010)
USA	Loam	Laboratory	n/a	Hardwood§	500	0, 10, 20, and 50 g kg^{-1}	No effect		Lim et al. (2016)
Colombia	Clay	Field	4 yr	Wood	500–700	0 and 20 Mg ha^{-1}	No effect		Major et al. (2012)
China	Loam	Plots	4 yr	Peanut shells	350–500	0 and 28 Mg ha^{-1}	No effect		Du et al. (2016)
Laos	Clay	Field	1 yr	Wood	n/a	0, 4, 8, and 16 Mg ha^{-1}	No effect		Asai et al. (2009)

† Switchgrass, *Panicum virgatum* L.; peanut, *Arachis hypogaea*; rice, *Oryza sativa*; corn, *Zea mays*.

‡ The numbers for Ksat within parentheses for Lim et al. (2016) are the SD of the mean.

§ The lowercase letters next to the mean values indicate significant differences.

¶ Only data for hardwood chip are included in this review.

n/a, not available.

ter infiltration, but high rates can reduce infiltration in sandy soils. Application of 0.5% biochar by weight (Ibrahim et al., 2013) and 25% biochar by volume (Githinji (2014) did not reduce infiltration but higher rates reduced infiltration. Similarly, saturated hydraulic conductivity linearly decreases in coarse-textured soils and linearly increases in fine-textured soils, with an increase in the amount of biochar applied (Table 7). Application of biochar at rates as low as 10 Mg ha^{-1} appears to either increase or decrease saturated hydraulic conductivity. Application of biochar at rates below 10 Mg ha^{-1} seems to have no significant effect on hydraulic conductivity.

The reduced infiltration and hydraulic conductivity resulting from biochar application in sandy soils can have positive implications for reducing nutrient leaching and increasing residence time and absorption of nutrients within the root zone (Kameyama et al., 2012). Rapid water flow in sandy soils increases concerns about leaching of nutrients and other chemicals, adversely affecting groundwater quality. Biochar could be a strategy to slow water flow in sandy soils and increase water flow in fine-textured soils. Increased water flow in soil with low infiltration rates or saturated hydraulic conductivity can be important for increasing water capture and storage and to reduce water losses in runoff. Overall, biochar application can generally decrease water infiltration and saturated and unsaturated hydraulic conductivity in sandy soils, and increase in clayey soils.

Water Retention and Plant Available Water

The effects of biochar on water retention appear to be more consistent than those on other soil hydraulic properties. A review of recent studies in Table 8 indicates that biochar increased water retention in 17 out of 19 soils, suggesting that biochar increases the ability of the soil to retain water in 90% of cases. Biochar particles can be highly effective at absorbing water. Table 8 also shows two trends. First, although various studies report increased water retention even at low rates of biochar application, two studies found that biochar only increased water retention when the amount applied was ≥15 Mg ha^{-1} (Paneque et al., 2016; Xiao et al., 2016). This suggests that large amounts of biochar can be required to increase water retention consistently. Second, two studies found no change or a decrease in water retention with biochar addition (Table 8). These two studies were conducted on clayey soils, which suggests that fine-textured soils may be less responsive to biochar application. Larger amounts of biochar may need to be applied to clayey than to sandy soils before changes are measurable.

The increased water retention can result in increased plant-available water, which is the difference in volumetric water con-

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Table 8. Impacts of biochar application on water retention for different soils and study types.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Water retention	Reference
					°C			
China	Sand	Laboratory	n/a	Wood	550	0, 7, 15, 25 g kg^{-1}	Increased	Zhang et al. 2016
Italy	Sand	Lysimeters	18 mo	Wood	~550	20 g kg^{-1}	Increased	Sorrenti and Toselli (2016)
Sri Lanka	Sand and sandy loam	Laboratory	6 mo	Rice husk	~600	0, 1, 5, and 10 g kg^{-1}	Increased	Gamage et al. (2016)
Denmark	Sand and sandy loam	Pot	3 mo	Straw and wood	750–1200	10 g kg^{-1}	Increased	Hansen et al. (2016)
Iran	Sandy loam	Laboratory	180 d	Rice husk and wood chip	350–550	20 g kg^{-1}	Increased	Esmaeelnejad et al. (2016)
Zambia	Sandy loam	Field	2 yr	Maize cob	350	0, 8, and 25 g kg^{-1}	Increased	Obia et al. (2016)
Spain	Sandy loam	Field	130 d	Wood and sewage sludge	500–620	0, 1.5, and 15 Mg ha^{-1}	15 Mg ha^{-1} Increased	Paneque et al. (2016)
Poland	Loamy sand	Pot	3 mo	*Miscanthus* × *giganteus* and wheat straw	300	0, 5, 10, 20, and 40 g kg^{-1}	Increased	Głab et al. (2016)
China	Loam	Field	4 yr	Peanut shells	350–500	0 and 28 Mg ha^{-1}	Increased	Du et al. (2016)
India	Loam	Laboratory	15 wk	Maize stalks	350	0, 5, 10, 20 g kg^{-1}	Increased	Kanthle et al. (2016)
China	Loam	Field	n/a	Crop straw	450	0 and 16 Mg ha^{-1}	Increased	Liu et al. (2016b)
China	Loam	Field	3 yr	Maize straw	400	0, 10, 20, and 30 Mg ha^{-1}	30 Mg ha^{-1} increased	Xiao et al. (2016)
Italy	Silty clay loam	Field	3 mo	Wheat bran	800–1200	14 Mg/ha	Increased	Andrenelli et al. (2016)
China	Clay loam	Field	3 yr	Maize straw and wood	n/a	0 and 7.8 Mg ha^{-1}	Increased	Ma et al. (2016)
China	Clay loam	Laboratory	180 d	Straw, wood, and wastewater sludge	500	0, 20, 40, and 60 g kg^{-1}	Increased	Zong et al. (2016)
Japan	Clay	Laboratory	180 d	Sugarcane bagasse	400–800	30 g kg^{-1}	No change	Kameyama et al. (2016)
Brazil	Clay	Field	3.5 yr	Woodchip	~450	0, 8, 16, and 32 Mg ha^{-1}	Decreased or no change	Carvalho et al. (2016)

† Rice, *Oryza sativa*; maize; *Zea mays*; wheat, *Triticum aestivum*; peanut, *Arachis hypogaea*; sugarcane, *Saccharum officinarum*

tent at -0.33 and -15 bar matric potentials. The review in Table 9 indicates that biochar application consistently increased plant-available water in 21 out of the 29 soils, suggesting that available water increases with biochar application in 72% of cases. Increase in plant-available water with biochar ranged from 4 to 130%.

The general increase in available water with biochar may have arisen from the increased specific surface area and porosity of biochar particles. The specific surface area ranges from 10 to 40 m^2 g^{-1} for sandy loam, 5 to 150 m^2 g^{-1} for silt loam, and 150 to 250 m^2 g^{-1} for clayey soils, but for biochar, it can be as high as 3000 m^2 g^{-1}. Biochar is a porous material and can thus retain water not only inside the pores but also between the particles because of the high specific surface area. In particular, micropores can hold water more strongly than macropores and mesopores via capillary and adhesive forces. Thus the addition of biochar to soil can alter the total porosity, pore size distribution, water transmission, and water retention characteristics (Table 7 to Table 9).

The increase in plant-available water with biochar suggests that the application of biochar to croplands could contribute to the reduction of the frequency of irrigation. This can be particularly important in water-limited or semiarid regions. The positive effect of biochar on increasing water retention can be larger in sandy soils than in clayey soils, as coarse-textured soils have lower microporosity and a smaller specific surface area than clayey soils. The summary of studies in Table 9 does not, however, show a clear trend for a larger increase in water retention in sandy soils compared with clayey soils. This can be partly attributed to the lack of paired studies specifically comparing different textural classes with the same amount of biochar being applied. The available studies were conducted in different soil textural classes, but with differing amounts of biochar being applied. Pyrolysis temperature, duration of experiment, and feedstock type varied among studies, making comparisons difficult. Most studies reporting improved water retention and plant-available water have used more than 1% of biochar, which is roughly equivalent to 25 Mg ha^{-1} of biochar. Smaller amounts of biochar may not significantly increase available water. For example, Głąb et al. (2016) and Xiao et al. (2016), and Kameyama et al. (2016) reported that biochar addition at 10 Mg ha^{-1} did not affect the available water content but higher rates of application significantly increased the available water. Most studies have been conducted in the laboratory and, in most cases, measurements were made immediately after mixing biochar with soil. Long-term field studies could better reflect soil–biochar interactions, which take time to develop.

Table 9 also shows some mixed effects of biochar on available water. First, biochar feedstock appears to influence available water. For example, Burrell et al. (2016) reported that straw biochar increased the available water but woodchip biochar had no effect under the same amount of biochar. Second, the seven studies showing no effects of biochar suggest that, in some cases, biochar may not increase available water regardless of the soil type

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Table 9. Effect of biochar application on plant-available water for different soil and management conditions.

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Plant-available water	Change	Reference
					°C		$cm^3\ cm^{-3}$	%	
					Increased				
Malaysia	Sand	Pot	170 d	Rice husk		0	0.047b‡		Manickam et al. (2015)
					550	20 g kg^{-1}	0.062ab		
						50 g kg^{-1}	0.082a	74	
Denmark	Coarse sand and sandy loam	Pot	5 wk	Wood and wheat straw	750–1200	0 and 10 g kg^{-1}	Increased 17–18	31–42	Hansen et al. (2016)
Zambia	Sandy loam, loamy sand, and sand	Field	1 yr	Corn cob and rice husk	350	0 to 40 g kg^{-1}	Increased		Obia et al. (2016)
Germany	Fine sand	Columns	200 d	Wood		0	0.157		Ajayi and Horn (2016)
						20 g kg^{-1}	0.188ns§		
						50 g kg^{-1}	0.195*	24	
					500–600	60 g kg^{-1}	0.248**	58	
	Silty clay loam					0	0.206		
						20 g kg^{-1}	0.215*	4	
						50 g kg^{-1}	0.232**	13	
						60 g kg^{-1}	0.269**	31	
Iran	Sandy loam	Incubation	180 d	Rice husk and wood	350–550	0 and 20 g kg^{-1}	Increased	≤130	Esmaeelnejad et al. (2016)
Brazil	Sandy loam	Field	3 yr	Wood	~450	0, 8, 16, and 32 Mg ha^{-1}	Increased		de Melo Carvalho et al. (2014)
USA	Sandy loam	Laboratory		Control		0 and 40 g kg^{-1}	0.18 b		Mollinedo et al. (2015)
				Corn stover	400–800	0.23 a	27		
				Switchgrass		0.23 a	27		
	Clay loam			Control		0.19 b			
				Corn stover		0.25 a	31		
				Switchgrass		0.23 a	21		
Italy	Silty clay loam	Field	3 mo			0	0.166b		Andrenelli et al. (2016)
				Wheat bran	800	14 Mg ha^{-1}	0.198a	19	
				Wheat bran	1200	0.182ab			
USA	Silty clay loam, loam, and sandy loam	Lab	2 wk	Switchgrass	375–475	0 and 1%	Increased		Koide et al. (2015)
China	Loam	Field	4 yr	Peanut shells	350–500	0 Mg ha^{-1}	0.21b		Du et al. (2016)
						28 Mg ha^{-1}	0.24a	14	
Poland	Loamy sand	Greenhouse	3 mo	*Miscanthus* × *giganteus* and winter wheat		5 g kg^{-1}	0.042c		Głab et al. (2016)
					300	10 g kg^{-1}	0.044c		
						20 g kg^{-1}	0.054b	29	
						40 g kg^{-1}	0.073q	74	

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Table 9. (cont.)

Location	Soil	Study type	Study duration	Biochar feedstock†	Biochar pyrolysis temperature	Biochar rate	Plant-available water	Change	Reference
					°C		$cm^3\ cm^{-3}$	%	
China	Clay loam	Field	3 yr	Corn straw and peanut hulls	n/a	0	0.106b		Ma et al. (2016)
						8 Mg ha^{-1}	0.137a	29	
China	Medium-textured	Field	3 yr	Corn residue		0	0.135b		Xiao et al. (2016)
					400	10 Mg ha^{-1}	0.136b		
						20 Mg ha^{-1}	0.139ab		
						30 Mg ha^{-1}	0.149a	10	
Japan	Clay	Incubation	180 d	Sugarcane		0	0.04c		Kameyama et al. (2016)
						10 g kg^{-1}	0.01bc		
					400–800	30 g kg^{-1}	0.06b	50	
						50 g kg^{-1}	0.06ab	50	
						100 g kg^{-1}	0.09a	125	
				Mixed Effect					
Austria	Sandy loam	Greenhouse	3 yr	Control		0			Burrell et al. (2016)
				Wheat straw		30 g kg^{-1}	Increased		
				Woodchip		30 g kg^{-1}	No effect		
				Vineyard waste	400–525	30 g kg^{-1}	Increased		
	Silt loam			Control		0			
				Woodchip		30 g kg^{-1}	No effect		
	Clay loam			Control		0			
				Woodchip		30 g kg^{-1}	No effect		
China	Sand	Incubation	180 d	Straw, woodchips, and wastewater sludge	500	20, 40, and 60 g kg^{-1}	No effect		Zong et al., 2016
China	Loam	Field	1 yr	Crop straw	450	0 to 16 Mg ha^{-1}	No effect		Liu et al. (2016b)
Ethiopia	Clayey	Field and lab	30 d	Corn stover and wood	450	5 g kg^{-1}	No effect		Bayabil et al. (2015)
Australia	Sandy loam	Field	30 mo	Wood	300–1000	0 and 47 Mg ha-1	No effect		Hardie et al. (2014)
Italy	Clay	Columns	2.5 yr	Wood		0	0.076ab		Castellinia et al. (2015)
					500	20 g kg^{-1}	0.085a		
						30 g kg^{-1}	0.065b		

* Significant at the 0.05 probability level.

** Significant at the 0.01 probability level.

† Rice, *Oryza sativa*; wheat, *Triticum aestivum*; corn, *Zea mays*; switchgrass, *Panicum virgatum* L.; peanut, *Arachis hypogaea*; sugarcane, *Saccharum officinarum*

‡ The lowercase letters for mean plant-available water values indicate significant differences.

§ ns, not significant

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and amount of biochar applied, which corroborates the site-specificity of biochar benefits (Table 9). Third, this review indicates that biochar may increase (Kameyama et al. (2016)), decrease (Castellinia et al., 2015), or have no effect (Bayabil et al., 2015) on available water in clayey soils. Coarse-textured soils appear to respond more consistently to biochar addition.

It is also important to mention that, in some cases, biochar application can increase water retention but this may not always result in increased plant-available water. In cases where biochar does not increase water retention, the hydrophobicity of biochar could play a role. As discussed earlier, fresh biochar produced at low temperatures can have water-repellent properties because of the presence of organic material on the surface of biochar particles. Overall, biochar addition can have positive effects on plant-available water content.

Soil Pore Size Distribution

Biochar application does not only increase total porosity (Table 1) but also alters the soil pore size distribution and the soil's intraaggregate structure. Pore size distribution has a greater effect on water, air, and heat flow relative to total porosity because energy flow within and through the soil depends more on the volume of pores within a specific pore size range than on the total volume of pores. Studies have found that biochar can reduce the proportion of drainable pores (Petersen et al., 2016) and increase proportion of mesopores (Lu et al., 2014)). The increase in mesopores with biochar application could be critical to increasing the ability of the soil to retain available water.

Medium-sized soil pores may be more important for the retention of available water than macropores and micropores. Large macropores drain rapidly and do not hold available water, whereas micropores hold water so tightly that much of the retained water may not be available to plants. Petersen et al. (2016) found that biochar converted drainable pores (60–300 μm) into water-retaining pores (0.2–60 μm). Quin et al. (2014) found that application of oil mallee (*Eucalyptus* spp.) biochar at 5% increased soil porosity and induced homogeneity in pore size distribution. Devereux et al. (2012) also reported that soil pore size decreased from 0.07 to 0.046 m^2 with 5% wood biochar application to a sandy loam and resulted in reduced saturated hydraulic conductivity and increased water retention capacity.

Biochar application can affect the pore size distribution of the soil through: (i) the inherently high porosity of biochar particles, (ii) the rearrangement of soil particles in contact with biochar particles, (iii) enhanced soil aggregation, and (iv) increased bioturbation (e.g., earthworm activity). Biochar can increase soil pore size distribution even in the short term because of porous nature of biochar particles (Andrenelli et al., 2016). Biochar particles can have a wider distribution of pore sizes than the soil (Sun and Lu, 2014). For example, the addition of biochar to soils with a narrow distribution of pore sizes such as sandy soils can rapidly increase the range of soil pore sizes, including macropores, mesopores, and micropores. An increase in the proportion of mesopores and micropores in sandy soils can increase their ability to retain water. It can also reduce rapid water infiltration, reducing loss of water from the root zone. The effect of biochar can be feedstock-specific. Using X-ray micro-computed tomography, Yu et al. (2016) characterized the soil pore structure and reported that woodchip and wastewater sludge biochar increased the porosity of soil macroaggregates more than straw biochar.

Similar to the effect on other soil physical properties, the effect of biochar on soil pore size distribution can be soil- and biochar-specific. For example, Hardie et al. (2014) found that application of 47 Mg ha^{-1} *Acacia* spp. green waste biochar did not affect the porosity of a sandy loam after 2.5 yr, but the proportion of macropores (>1200 μm) increased because of earthworm burrowing in biochar-amended soils. Some studies have shown that earthworms ingest biochar-soil particles and translocate biochar to greater depths in the soil profile through burrowing or bioturbation (Lehmann et al., 2011). Overall, biochar addition alters the soil pore size distribution and soil structural characteristics, but the extent of the changes could depend on biochar type (e.g., feedstock, hydrophobicity) and soil.

BIOCHAR AND SOIL THERMAL PROPERTIES

Soil temperature and thermal properties are primary factors influencing energy balance as well as soil's physical, chemical, and biological processes including water and gas fluxes, aeration, evaporation, drainage, chemical reactions (e.g., oxidation, reduction), organic matter decomposition, microbial activity, seed germination, and others. Changes in soil temperature and thermal properties resulting from biochar amendment can be critical for understanding the soil response to climatic fluctuations and energy balance in the soil–atmosphere interface.

Soil Temperature and Albedo

Biochar application could alter soil temperature because of its intrinsic electrical and thermal properties as well as by exerting changes in soil properties. The few published studies on this topic have found that biochar application may affect soil temperature. On a winter wheat–corn field in the North China Plain, corncob biochar application at 4.5 and 9.0 Mg ha^{-1} reduced daytime soil temperature fluctuations (Zhang et al., 2013). The same study found that biochar reduced soil temperature by 0.8°C in high-temperature soils and increased the soil temperature by as much as 0.6°C in low-temperature soils. On grassland and fallow fields in Poland, wood biochar application reduced the amplitude of daily soil temperature in a grassland and increased the amplitude in fallow soil, but the mean soil temperature measured at the 0- to 15-cm depth did not differ (Usowicz et al., 2016). In Italy, Ventura et al. (2012) found no effect of biochar application at 30 and 60 Mg ha^{-1} on soil temperature at 7.5 cm depth. These findings suggest that biochar could reduce daytime soil temperature and increase nighttime soil temperature, although mean soil temperature may not differ significantly. These changes in soil temperature could have significant implications for reducing extreme temperature fluctuations in the soil.

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The dark color of biochar can darken the soil and change the albedo of the soil surface. Albedo is the proportion of solar radiation or light reflected by the soil back to the atmosphere. Dark soils can absorb more sunlight and have lower surface reflectance than grayish or pale soils. Studies in China (Zhang et al., 2013) and Poland (Usowicz et al., 2016) have reported that biochar addition at rates ranging between 4.5 and 30 Mg ha^{-1} reduced albedo or the reflectance of the soil. Soils with biochar can have lower reflectance than soil without biochar, reducing evaporation and increasing soil water content. Verheijen et al. (2013) found that surface application of pine (*Pinus* spp.) biochar at rates of 1, 10, 50, 100, and 200 Mg ha^{-1} can reduce albedo of the soil surface even at low application rates. Changes in soil albedo can be more notable in degraded soils with limited or no vegetation cover than in soils with abundant vegetation cover, which minimizes differences in albedo (Usowicz et al., 2016).

Soil Thermal Properties

Soil thermal properties include thermal conductivity, volumetric heat capacity, and thermal diffusivity. These properties directly influence the energy balance and the storage and transfer of heat through the soil. Zhang et al. (2013) found that corncob biochar addition reduced soil thermal conductivity by 3.5% under 4.5 Mg ha^{-1} of biochar and by 7.5% under 9 Mg ha^{-1} of biochar. Similarly, Usowicz et al. (2016) found that biochar application at 10, 20, and 30 Mg ha^{-1} reduced thermal conductivity as well as thermal diffusivity in the 0- to 10-cm soil depth. In addition, Zhao et al. (2016) observed that soil thermal conductivity and thermal diffusivity decreased with biochar application at 4.5 and 9.0 Mg ha^{-1}.

These few studies on both soil temperature and thermal properties indicate the following points. First, biochar application can moderate soil temperature extremes or regulate abrupt fluctuations in soil temperature. Second, biochar application can consistently reduce the amount of heat transferred through the soil (known as thermal conductivity). The temperature-regulating potential of biochar is promising and suggests that biochar amendment can be strategic for reducing and managing increasing climatic fluctuations such as heat waves and droughts. The addition of biochar can also change the soil reflectance by altering the soil surface color. Change in soil temperature appears small and may depend on the rate of biochar application. Higher rates (>30 Mg ha^{-1}) of biochar than those reported by Zhang et al. (2013) and Usowicz et al. (2016) may be needed to reduce soil temperature significantly. Soil thermal conductivity decreases after biochar application and appears to be more sensitive to biochar than soil temperature.

Soil thermal properties are significantly correlated with other soil properties including bulk density, water content, air-filled porosity, organic matter concentration, solute concentration aggregate stability, particle size, and mineralogy, among others. Thus changes in these soil properties after biochar application can concurrently alter the soils thermal properties. For example, soil thermal conductivity can increase with an increase in bulk density and soil water content and decreases with an increase in soil organic matter and salt concentration (Abu-Hamdeh and Reeder, 2000; Zhao et al., 2016). The decrease in bulk density with biochar application (Table 1) can reduce the thermal conductivity, whereas the increase in water retention with biochar application (Table 8) can increase the thermal conductivity. For example, Zhang et al. (2013) found that soil thermal conductivity decreased as soil bulk density decreased with increasing amounts of biochar application. A decrease in bulk density caused by biochar addition directly increases porosity. The increased porosity can result in greater air-filled pore spaces and fewer points of contact or lower connectivity among soil particles, thereby reducing heat transfer and thermal conductivity. The thermal conductivity of the air is much lower than that of water, soil, and biochar (Usowicz et al., 2016).

The relationship of thermal conductivity with biochar-induced changes in soil properties can be complex because of interacting effects of soil properties. Usowicz et al. (2016) indicated that the rate of thermal conductivity increased as soil water content increased but the thermal conductivity increase was smaller in soil with lower rather than higher bulk density. Management can also affect the strength of any correlation of thermal properties with other soil properties after biochar application. For example, Usowicz et al. (2016) reported that biochar application had a greater effect on thermal properties in fallow land than in grassland in a temperate climate. The literature review here indicates that biochar application reduces soil thermal properties and moderates soil temperature. It also indicates that biochar can impact soil thermal properties by altering other soil properties (bulk density, porosity, water content, and others). The extent of changes in soil thermal properties could depend on the extent of changes in other soil properties after biochar application.

FACTORS INFLUENCING BIOCHAR EFFECTS ON SOIL PHYSICAL PROPERTIES

Figure 1 shows some factors that can affect the extent to which biochar can alter soil physical properties. It is important to note that the factors do not act alone but interact with other factors. For example, the amount of biochar required to improve soil properties could depend on the soil type as well as climate. Most studies have evaluated two factors, including biochar application rates and soil textural classes, but studies on other factors are few.

Biochar Amount

One of the primary factors that influence biochar effect on soil physical properties is the amount of biochar applied. According to the literature reviewed, as the amount of biochar applied increased, larger changes in the soil properties occurred. For example, in general, bulk density (Table 1) decreases, wet aggregate stability (Table 3) increases, water repellency (Table 6) increases, and water retention (Table 8) increases as the amount of biochar applied increases. The decrease or increase with an increase in the amount of biochar applied often follows a linear function, but in a few cases, the relationship could be quadratic, suggesting that the effect of biochar at larger rates could be smaller or have no effect (Rogovska et al., 2016). However, some soil physical properties (e.g., penetration resistance) do not appear to change readily with the amount of biochar used (Table 2). Kameyama et al. (2012)

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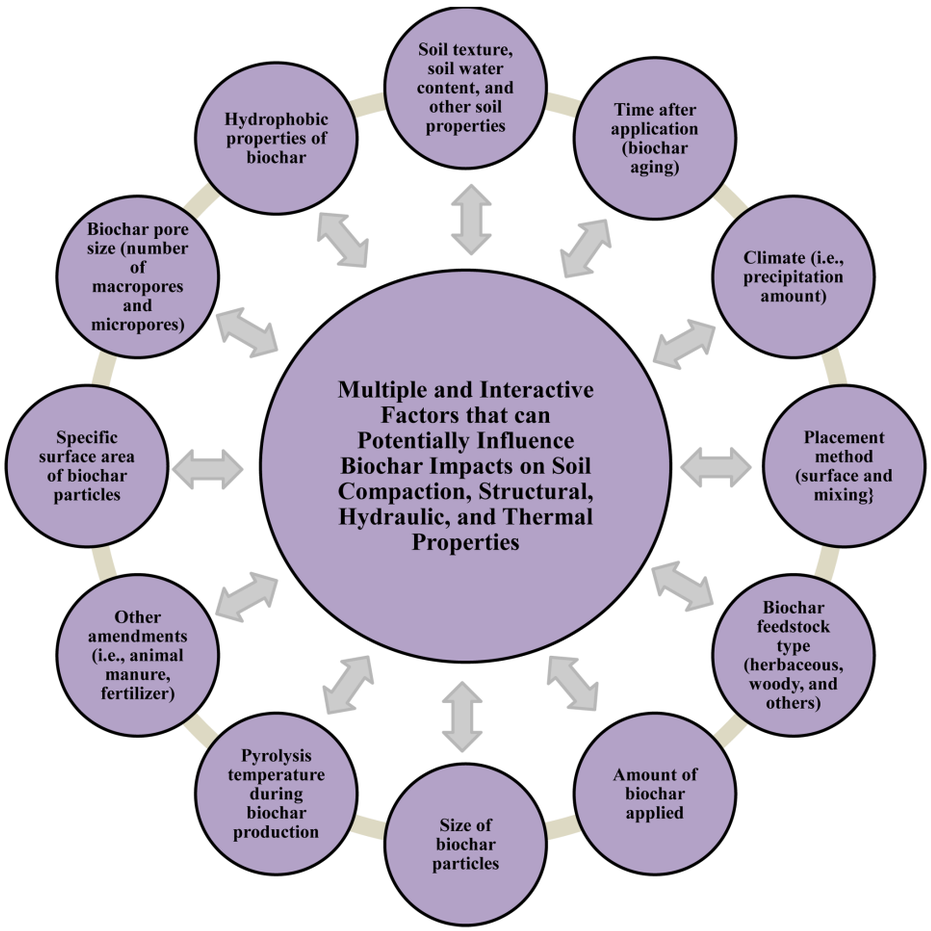


Fig. 1. Factors including soil properties, biochar properties, and management scenarios and their interactions affect the impact of biochar application on soil physical properties.

found 1 to 3% biochar application increased unsaturated hydraulic conductivity only in a matric potential range of more than -10 kPa, suggesting that higher rates of biochar application may be required to significantly increase unsaturated flow in clayey soils than in sandy soils. In some cases, however, even high rates of biochar application appear to have inconsistent effects on hydraulic conductivity. On a loamy sand, Hardie et al. (2014) reported that biochar at 47 Mg ha^{-1} increased unsaturated hydraulic conductivity at -0.25 and -0.10 kPa but had no effect at other water potentials. This suggests that larger amounts of biochar may be needed to make significant changes in some soil physical properties.

Soil Texture

Soil particle size distribution can be an important factor affecting biochar impacts on soil properties. In general, biochar appears to have greater benefits on coarse-textured soils than on fine-textured soils (Omondi et al., 2016). For example, the extent of a decrease in bulk density and an increase in water retention capacity with biochar application can be larger in coarse-textured than in fine-textured soils (Tables 1 and 8). This could be caused by the larger differences in size and density between biochar and sand particles being larger than those between biochar and clay particles. Small biochar particles can readily fill in the large pore spaces in sandy soils and increase tortuosity for water flow, reducing saturated hydraulic conductivity relative to clayey soils. Furthermore, biochar particles may provide organic binding agents to sandy soils, improving soil aggregation to a greater degree than in clayey soils.

Amendments

A combination of biochar and inorganic fertilizer or other organic amendments (e.g., animal manure) could improve soil properties more than biochar alone. For example, biochar combined with animal manure could increase soil aggregation compared with biochar alone as manure could provide labile or transient organic binding agents to bind biochar and inorganic particles into stable aggregates (Khademalrasoul et al., 2014). Biochar could also improve the efficiency of N fertilizer for crop production (Chan et al., 2007), but the effects of biochar × N fertilizer interactions on soil physical properties have not been studied in depth. Most previous studies on soil physical properties have evaluated the benefits of biochar alone and not the combination of biochar and other amendments. Combining biochar with other amendments could improve the effectiveness of biochar for improving soil physical properties (Lentz et al., 2014).

Placement Method

Biochar benefits can depend on the biochar placement method. The published field studies indicate that biochar is commonly incorporated into the soil to reduce losses from wind and water erosion as well as to improve biochar-soil contact. A few studies have compared biochar performance between surface application and incorporation. Mixing of biochar with soil reduced soil water repellency (Page-Dumroese et al., 2015) and bulk density (Usowicz et al., 2016) compared with surface application. However, Verheijen et al. (2013) reported that surface application of biochar can have more positive effects on reducing soil surface albedo than biochar incorporated into the soil. Reducing surface albedo can moderate climate. Application of biochar to no-till systems without incorporation could reduce surface albedo but may have smaller effects on soil physical properties than biochar mixed with tillage. Studies comparing biochar performance under different biochar placement methods or tillage systems are limited.

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Table 10. Summary of the impacts of biochar application on different soil physical properties.

Soil physical properties	Biochar effect	Number of study soils
	Does biochar application affect compaction and mechanical properties?	
Bulk density	Decreases by 3–31%	>22†
Particle density	Decreases by 14–64%	2
Total porosity	Increases by 2–41%	>14†
Penetration resistance	May or may not decrease	6
Tensile strength	Decreases by 42–242%	4
Liquid limit	Increases by 8–22%	2
Plastic limit	Mixed effects	2
Plasticity index	Increases or no effect	2
	Does biochar application improve soil structural properties?	
Percentage of water-stable aggregates	Generally increases by 21–226%	13
Mean weight diameter of water-stable aggregates	Generally increases by 4–58%	21
Dry aggregate stability	Increases or has no effects	8
	Does biochar application alter water transmission characteristics?	
Soil water repellency	Small and mixed effects	5
Water infiltration	Reduces in sandy soils and increases in clayey soils	6
Saturated hydraulic conductivity	Decreases in coarse-textured soils by 7–2270% and increases in fine-textured soils by 25–328%.	28
Unsaturated hydraulic conductivity	Reduces in sandy soils and increases in clayey soils	3
	Does biochar application enhance water retention capacity?	
Water retention	Increases in most soils (90%) but not clayey soils	19
Plant-available water	Increases in most soils (72%) from 4–130%	29
	Does biochar application change soil thermal properties?	
Soil temperature	Reduces daytime soil temperature and increases nighttime temperature	3
Soil thermal conductivity	Reduces soil thermal conductivity	3

† This review reports data on bulk density and porosity only from studies published in 2016. Omondi et al. (2016) reviewed studies prior to 2016.

Time after Application

The effect of biochar on soil physical properties could change with the time after application. For example, Briggs et al. (2012) found water repellency was higher for fresh wooden biochar than old biochar. Oxidation and drying of biochar with time can reduce organic compounds on the surface of biochar particles, reducing hydrophobic properties. Also, biochar C can interact with soil organic and inorganic particles as time passes, improving soil aggregation and other processes (Briggs et al., 2012). As biochar ages, physical, chemical, and biological reactions on the surface of biochar particles can alter their properties, which can, in turn, influence biochar effects on soil physical properties. Monitoring changes in soil physical properties with time following biochar application is warranted to ascertain how biochar effects change with time.

Biochar Particle Size

Biochar particle size can directly affect biochar–soil interactions, influencing changes in soil physical properties. Small biochar particles could more easily mix or interact with soil particles to form aggregates than large biochar particles (Herath et al., 2013). In addition, the smaller the biochar particles, the greater the specific surface area per unit of mass. For example, water retention increases with an increase in the total specific surface area per unit of mass. Głąb et al. (2016) found that bulk density decreased, total porosity increased, available water content decreased, and water repellency decreased with an increase in biochar size from 500 to 2000 μm. Because small biochar particles often have more micropores than large biochar particles, they could hold water more strongly and have greater available water content than large particles. However, small particles could reduce saturated water flow in the soil by clogging the pore space. Esmaeelnejad et al. (2016) observed that biochar application reduced saturated hydraulic conductivity as biochar particle size decreased. Not only the porosity of the biochar particles affects soil properties but also the arrangement of biochar and soil particles. Biochar particle size could strongly influence changes in soil physical properties.

Feedstock Type

Published literature concerning biochar feedstocks indicates the following points. First, feedstock type can have different effects on soil physical properties. Second, even within herbaceous or wood feedstock groups, biochar could have different impacts. For example, within herbaceous feedstocks, Głąb et al. (2016) found that water content at field content and plant-available water content were greater for *Miscanthus* × *giganteus* J.M.Greef , Deuter ex Hodk., Renvoize biochar than for wheat straw biochar but other soil physical properties did not differ. Gray et al. (2014), within woody feedstocks, reported that Douglas fir [*Pseudotsuga menziesii* (Mirb.) Franco] biochar adsorbed more water than hazelnut (*Corylus* spp.) shell biochar because the former had greater porosity. The few reports comparing herbaceous and woody feedstock effects on soil physical properties suggest that herbaceous feedstocks can have more beneficial effects on soil physical properties than woody feedstocks. Burrell et al. (2016) found that wheat straw reduced bulk density and increased the amount of water-stable aggregates and available water more than wood feedstocks Similarly, crop straw biochar (Sun and Lu, 2014) and bamboo (*Bambusa* spp.) biochar (Demisie et al., 2014) increased wet soil aggregate stability but wood biochar had no effect. Though herbaceous feedstocks appear to enhance soil physical properties more than woody biochar, more comparisons among

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biochar feedstocks are needed to discern the different impact of biochar (Esmaeelnejad et al., 2016).

Pyrolysis Temperature

The temperature used during the thermochemical conversion of biochar can influence the extent to which biochar affects soil properties. Biochar produced at high temperatures (≥500°C) could, in general, retain more water and have lower water repellency than those produced at low temperatures (Kinney et al., 2012; Gray et al., 2014). The lower water repellency at high temperatures can be attributed to the removal of the hydrophobic organic materials coating the surface of biochar particles. Kinney et al. (2012) reported that the water repellency of biochar decreased 13-fold when produced at 500°C compared with when it was produced at 300°C. In some cases, the pore space of biochar can increase with an increase in pyrolysis temperature, as charring can unclog pores and reduce hydrophobicity. Porous biochar particles can capture and conduct water more readily. Some have found that differences in the pyrolysis temperature of biochar may have small effects on soil physical properties (Herath et al., 2013; Andrenelli et al., 2016; Burrell et al., 2016; Esmaeelnejad et al., 2016).

RESEARCH NEEDS

1. Field studies on biochar are few. The majority of studies have been conducted in laboratory or greenhouse conditions. The lack of field research data hinders our ability to conclusively ascertain the positive and negatives impacts of biochar on soil physical properties. The performance of biochar between small greenhouse or laboratory pots and large fields with variable soil and environmental conditions could differ. For example, representative measurements of compaction parameters (bulk density and penetration resistance), water infiltration, water erosion, and soil temperature fluctuations require field experiments. Thus more field studies are needed to better characterize biochar application impacts and recommend large-scale application and use of biochar.
2. Published information is mostly from short-term laboratory or greenhouse studies (<2 yr). Long-term studies are needed to verify the promising laboratory results under field conditions. Long-term studies can allow time for biochar to react and interact with soil particles. Soil responses, particularly the responses of soil physical properties, to biochar application can be slow. The extent of the benefits of biochar may not be fully realized until biochar–soil interactions occur in the long term.
3. Studies of biochar under different soil conditions (e.g., differing soil organic C levels and soil texture), biochar management (e.g., biochar age or longevity; feedstock type), combination with other amendments (e.g., animal manure or inorganic fertilizers), field management (e.g., tillage or crop rotations) are lacking. However, this information is needed to draw conclusions and make practical recommendations for large-scale use of biochar for different management and climatic scenarios.
4. Most studies have evaluated select soil physical properties such as bulk density, water retention, saturated hydraulic conductivity, and wet aggregate stability (Table 10). Data on other equally relevant properties such as mechanical, rheological, and thermal properties are scanty. More comprehensive and site-specific studies assessing all soil properties and soil services are warranted.
5. The minimum threshold level of application at which biochar improves soil properties and the maximum threshold level at which biochar no longer improves soil properties should be determined for different soils and management scenarios. This information is needed to evaluate the economic viability of this technology.
6. Most previous studies have assessed changes in soil properties only for biochar incorporation to about 15 cm depth. The potential of biochar for improving soil properties at greater depths is yet to be determined. Biochar-induced changes in soil properties in the deeper soil profile could be critical for retaining or storing water and nutrients for plant growth.
7. Further studies are needed to determine the mechanisms by which biochar affects soil physical properties. In particular, the effect of biochar properties (feedstock type, particle size, pyrolysis temperature, and others) on soil physical properties appears to be mixed. The need exists to focus less on effects and more on the mechanisms by which biochar application alters soil physical properties.
8. There is limited research on biochar benefits in degraded or problem soils (low organic matter, low fertility, eroded, compacted, saline, saline-sodic, sodic soils, and others). The physical properties in such soils could benefit more from biochar application than highly fertile or productive soils. Field experiments are needed to confirm this hypothesis.
9. More studies comparing biochar effects under different placement methods or tillage systems are needed. These comparisons require field studies to determine how large-scale application of biochar can perform under different application methods. Placement method could affect the extent to which biochar particles interact with soil particles and create the changes in soil properties.
10. While biochar application may provide positive benefits to soil properties, the economics of biochar application and use large scales should also be reviewed and fully discussed to provide practical recommendations.

SUMMARY AND CONCLUSIONS

The literature, in general, suggests that biochar application can exert changes in most soil physical properties with the extent depending on the soil property. The extent of any impacts appears to be biochar-, soil-, and management-specific. Based on Table 10, the following can be concluded regarding biochar effects on soil physical quality.

- Biochar application could reduce the risks of soil compaction, but field data on penetration resistance are very limited to make definite conclusions. In general,

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biochar reduces soil bulk density and tensile strength (the soil's resistance to rupture) and increases porosity (Table 10). The consistent decrease in tensile strength indicates that biochar-amended soils can be more friable and less compactible than soils without biochar. Biochar also improves soil consistency and reduces cohesiveness.

- Biochar application generally improves soil's structural quality. Biochar can improve soil aggregation and increase the proportion of water-stable aggregates. This can result in increased porosity and water infiltration, and reduced the risks of water erosion. However, biochar may or may not increase dry soil aggregate stability, which is an indicator of wind erosion potential. Additional data are needed to understand if biochar application can reduce wind erosion risks.
- Biochar application can alter water transmission characteristics in the soil. Biochar seems to have little or no effect on water repellency, but biochar strongly alters saturated hydraulic conductivity. Biochar application reduces saturated water flow in coarse-textured soils and increases it in fine-textured soils. The trend is similar for water infiltration although field data on infiltration are limited. Saturated hydraulic conductivity is often measured in the lab on small cores, whereas water infiltration is measured in the field. Because biochar has been mostly studied in laboratory or greenhouse settings, more data on saturated hydraulic conductivity are available than for water infiltration. Similar to saturated hydraulic conductivity, biochar can reduce unsaturated hydraulic conductivity in sandy soils and increase it in clayey soils.
- Studies on water retention are abundant and indicate that biochar application can consistently increase water retention capacity and increase the amount of plant-available water. Biochar appears to have the most consistent effect on water retention of all soil physical properties. About 90% of the recent biochar studies report increased water retention. The increase in plant-available water with biochar can have important applications for soil water management, particularly in water-limited regions.
- Studies on soil thermal properties are few but suggest that biochar application could moderate soil temperature and reduce abrupt fluctuations in soil temperature. The beneficial effects of biochar on soil thermal properties can have large implications for the storage and transfer of heat through the soil and mitigation of climatic fluctuations.

Not all the factors that influence biochar effects on soil properties have been studied in depth (Fig. 1). For example, few have compared the effect of feedstock type, pyrolysis temperature, biochar particle size, and time after biochar application, under the same amount of biochar applied but have suggested that such factors may significantly affect the extent to which biochar impacts soil physical properties. In addition, mechanisms by which biochar alters soil physical processes (e.g., aggregation) are not well understood. The data available are primarily from short-term and laboratory studies. More field studies under different soil types, management scenarios, and climates are needed to better understand how different types of biochar perform under field conditions. Furthermore, most previous studies have measured select physical properties. More comprehensive assessments of all soil physical properties under long-term field studies are warranted to fully discern biochar–soil interactions over time. Overall, the review suggests that the use of biochar could reduce the risks of soil compaction and improve soil structural, hydraulic, and thermal properties but more field-scale data are needed to further improve our understanding of biochar potential as an amendment to manage soils.

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